Politecnico di Milano

Biochip A.Y. 2021/2022 - M. Carminati

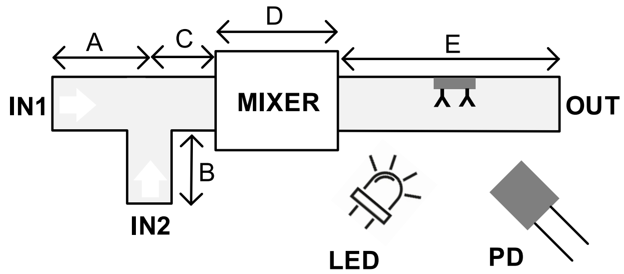
Exam of 23.06.2023

Test duration: 2 hours – No support material (textbooks, laptops, smartphones, notes) admitted.
All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.

Exercise 1

Consider the following microfluidic device used to detect target molecules by means of **fluorescence**. Molecules enter from port IN1, while fluorophores (GFP) enter from port IN2 for labeling. They mix in the mixer and then pass in the detection chamber where the sensing area (in the middle of section E) is functionalized with receptors (Y). The height h and width w of all the channels is constant.

(I) Device (Top View)



Data:

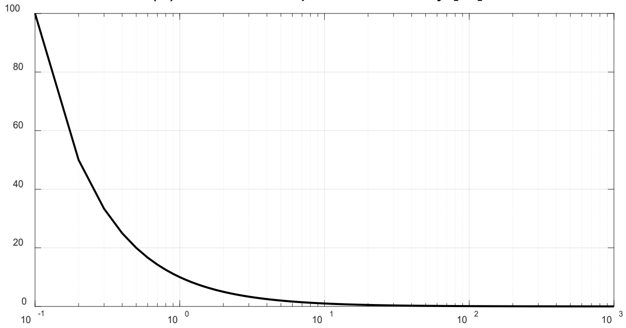
h = 30 μm
w = 50 μm

A = 2 mm
B = 1 mm
C = 1 mm
D = 4 mm
E = 10 mm

1atm = 101 kPa
$\eta_{water} = 10^{-3}$ Pa·s

a) Draw the complete electrical fluidic equivalent of the system (Fig. I), considering a fluidic resistance of $R_{MIX} = 2 \cdot 10^6$ [Pa·s/m^3] for the **mixer**.
b) Draw the scheme of a **Tesla mixer** with 4 stages. Is the structure symmetrical in terms of fluidic resistance?
c) Assuming an open outlet and $P_{IN1} = P_{IN2}$ = 3 atm, find the flow rate in all the sections.
d) In order to achieve a molecule capture efficiency of the detector better than 60% (Fig. II), choose the proper **pressure drop** needed across segment E.

(II) Molecule Capture Efficiency [%]



e) Describe possible reasons that could originate a **fluidic capacitance**.
f) Discuss the difference (pros and cons) between sandwich and competitive assays. To which category does this system belong to? Could labeled detection be replaced by **label-free** optical detection?
g) Which key optical element is missing for fluorescent detection in the scheme of Fig. I?

Exercise 2

Now focus on the circuit to optically detect the binding events from the photodiode current signal. Assume a max. current of 1 mA and a photodiode capacitance of 10 pF.

Data: V_{DD} = +5 V, V_{SS} = -5 V OpAmp equivalent noise: e_n = 2 nV/sqrt(Hz) i_n = 10 fA/sqrt(Hz)

a) Draw the scheme of a current amplifier to sense the current of the photodetector. Properly size the components of the circuit.
b) Considering only the noise of the feedback resistor of the TIA, choose the TIA bandwidth in order to have a current resolution of 400 pA RMS. How can this bandwidth be achieved?
c) Consider now **all the noise sources**, and plot the input referred total power spectral density of current noise.
d) Consider now the **shot noise** associated with the photodetector current: what would be its impact in the worst case?
e) Propose a solution to implement a **lock-in** architecture (modulation and demodulation) on the optical readout system.
f) If now the optical detection is replaced by an **electrochemical detection** approach, propose a solution to detect the binding event based on the same TIA topology designed in (a).
g) In order to **thermally release** the molecule after binding, should the detector temperature be increased or decreased? (motivate the answer).

Multichance students with 30% reduction of the exercises please skip points f) and g) of both exercises.